**DEPARTMENT OF ELECTRONICS ENGINEERING**
**INDIAN INSTITUTE OF TECHNOLOGY (ISM) DHANBAD**
**SUBJECT: ECC11101 ELECTRONICS ENGINEERING (3-0-0)**
**CLASS: 1st SEMESTER B.Tech. WINTER SEMESTER 2018-2019**

## LECTURE PLAN

| Sl No. | Topics | No. of Lectures |
|---|---|---|
| 1. | Introduction to Electronic Engineering course | 1 |
| | **SEMICONDUCTOR MATERIALS** | |
| 2. | Semiconductor materials and their properties,<br>Charge carriers in solids – covalent bond, hole and band gap energy<br>Modification of carrier density – intrinsic and extrinsic semiconductor, Majority and minority carriers<br>Transport of carriers – drift, diffusion, mobility, conductivity, Einstein relationship | 4 |
| | **PN JUNCTION** | |
| 3. | pn junction formation, pn junction in equilibrium, charge concentration, electric field in pn junction, built-in potential<br>pn junction behavior – open circuit, forward bias and reverse bias, junction capacitance under reverse bias<br>I/V characteristics, constant voltage model, diode model, reverse breakdown – zener and avalanche breakdown | 5 |
| | **DIODE APPLICATIONS** | |
| 4. | Diode – ideal, exponential and constant voltage models<br>half wave and full wave rectifier – operation, input output waveform, peak voltage and average voltage, capacitor filter, ripple factor<br>diode clippers – top clipper, bottom clipper and slicer<br>diode clamper – bottom clamping, top clamping<br>voltage multipliers<br>zener diode use as voltage regulators | 5 |
| | **BIPOLAR JUNCTION TRANSISTORS** | |
| 5. | Structure of bipolar junction transistor (BJT), operation and dc behavior of BJT<br>BJT small signal equivalent circuits, Characterizing BJT amplifiers (CE, CB and CC)<br>Basic amplifier circuit, DC biasing, Q point, load line and transfer characteristics<br>Fixed bias circuit, bias stability, voltage divider bias<br>CE amplifier frequency response (low frequency, mid frequency and high frequency) | 11 |
| | **FIELD EFFECT TRANSISTORS** | |
| 6. | Structure of field effect transistor (FET), operation of FET, Pinch off voltage, V-I characteristics, small signal equivalent circuit, common source amplifier | 3 |
| | **OPERATIONAL AMPLIFIER** | |
| 7. | Ideal Operational amplifier<br>Linear applications – inverting and non-inverting amplifier, unity gain buffer, summer, Difference amplifier, instrumentation amplifier, integrator and differentiator, low pass filter<br>nonlinear applications of op amp – comparator, Schmitt trigger and Square wave oscillators | 5 |

| DIGITALCIRCUITS | | |
|---|---|---|
| 11. | Number systems –decimal, binary, octal, hexadecimal, complement of a number, addition, subtractions using n and n-1 complement method<br>Boolean algebra, SOP and POS, simplification of logic functions using Boolean theorems and Karnaugh map (4 variable)<br>Digital gates – OR, AND, NOT, NAND, NOR, XOR, XNOR<br>Implementation of logic functions using logic gates, Half added, full adder, Introduction to sequential circuits | 8 |
| | TOTAL | 42 |

**Reference Books:**


1. Boylestad, R L and Nashelsky L, Electronic Devices and Circuit Theory. 11th Ed., Pearson 2013
2. J. Millman, C Halkias and C Parikh Integrated Electronics, Analog and digital Circuits and systems
3. Leach & Malvino Digital Principles and Application


Mid Semester Examination will cover the course up to 15 Feb. 2019